AP PHYSICS 1

Unit 1 • Lesson 1.2 • Notes (filled in)

Topic 1.2 • Displacement, Velocity, and Acceleration

Notes (filled in)

Objective: define velocity and acceleration, and use the kinematic equations for constant acceleration.

Definitions	Average velocity: $v_{\text{avg}} =$ $\frac{\Delta x}{\Delta t}$ Average speed $=$ $\frac{\text{distance}}{\Delta t}$ Average acceleration: $a_{\text{avg}} =$ $\frac{\Delta v}{\Delta t}$ Units: v in m/s, a in m/s^2 Acceleration is the rate of change of velocity: how much the velocity changes each second.
Speeding up or slowing down?	v and a have the **same** sign: the object is speeding up. v and a have **opposite** signs: the object is slowing down. A negative acceleration does not always mean slowing down: a car moving in the $-x$ direction with $a < 0$ speeds up.
The kinematic equations (constant a only)	1) $v =$ $v_0 + at$ no Δx 2) $\Delta x =$ $v_0 t + \frac{1}{2}at^2$ no v 3) $v^2 =$ $v_0^2 + 2a\,\Delta x$ no t Strategy: list the known quantities (v_0, v, a, t, Δx), find the one that is not needed, pick the equation without it.
Example 1: a car speeds up	A car goes from 4 m/s to 16 m/s in 3 s with constant acceleration. $a = \frac{\Delta v}{\Delta t} =$ $\frac{16-4}{3} = 4\,\text{m/s}^2$ $\Delta x = v_0 t + \frac{1}{2}at^2 =$ $12 + 18 = 30\,\text{m}$
Example 2: braking	A car moving at 20 m/s brakes with $a = -5\,\text{m/s}^2$ until it stops. Time not given: use equation 3. $0 = 20^2 + 2(-5)\Delta x \Rightarrow \Delta x =$ 40 m Time to stop: $t =$ $\frac{0-20}{-5} = 4\,\text{s}$
Free fall	Near Earth, an object in free fall (no air resistance) has $a =$ $g \approx 10\,\text{m/s}^2$ downward, whether it moves up or down. A stone is dropped from rest. After 2 s: $v =$ 20 m/s downward; it has fallen $\frac{1}{2}(10)(2)^2 = 20\,\text{m}$.
Summary	$v_{\text{avg}} = \Delta x/\Delta t$ and $a = \Delta v/\Delta t$. Same signs: speeding up; opposite signs: slowing down. With constant acceleration use $v = v_0 + at$, $\Delta x = v_0 t + \frac{1}{2}at^2$, $v^2 = v_0^2 + 2a\Delta x$. In free fall $a = g = 10\,\text{m/s}^2$ downward.